

Class 10 NCERT Maths

CLASS 10 NCERT MATHS

Chapter 14: Probability (Core Basics Edition)

CHAPTER 14: PROBABILITY

A simplified, detailed guide in Hinglish

Special edition for students who skipped Class 7-8-9

IS PDF MEIN RANG KA MATLAB:

RED text = core basic / prerequisite (Class 7-8-9 ka).

GREEN text = exam mein BAAR-BAAR aata hai (zaroor yaad rakh).

(Aur black = normal explanation.)

CORE BASICS - MISS MAT KARNA (RED PAGE)

1) PROBABILITY KYA HAI?

Probability = kisi cheez ke hone ki sambhavna (chance). 0 se 1 ke beech hoti hai.

2) EXPERIMENT, OUTCOME, EVENT

Experiment = jo karte ho (coin uchhalna).

Outcome = ek result (Head ya Tail).

Event = outcomes ka group (jaise "even number aaye").

3) FAVOURABLE OUTCOMES

Favourable = jo hum chahte hain wale outcomes ki ginti.

4) TOTAL OUTCOMES (saare possible)

Coin: 2 (H,T). Dice: 6 (1-6).

Deck of cards: 52.

5) FRACTION

Probability ek fraction hota hai:

favourable / total. (Jaise $3/6 = 1/2$)

6) DECK OF CARDS (52)

4 suits: Hearts, Diamonds (laal); Spades, Clubs (kaali).

Har suit 13 cards. Face cards = J,Q,K (12 total).

Aces = 4.

7) SURE vs IMPOSSIBLE

Sure event -> probability 1. Impossible -> 0.

NOW: ASLI CHAPTER 14 SHURU

TOPIC 1: PROBABILITY FORMULA

EXAM ALERT: Sabse important formula:

$$P(E) = \frac{\text{(Number of favourable outcomes)}}{\text{(Total number of outcomes)}}$$

$P(E)$ hamesha 0 aur 1 ke beech ($0 \leq P(E) \leq 1$).

TOPIC 2: COMPLEMENT (not E)

EXAM ALERT: "E nahi hone" ki probability:

$$P(\text{not } E) = 1 - P(E)$$

(Dono ka sum hamesha 1 hota hai.)

TOPIC 3: SURE AUR IMPOSSIBLE EVENTS

Sure event (pakka hoga) $\rightarrow P = 1$.

Impossible event (kabhi nahi) $\rightarrow P = 0$.

[CORE] Probability kabhi 1 se zyada ya 0 se kam nahi ho sakti.

SOLVED EXAMPLES: HARDEST -> EASIEST

8 examples पूरी तरह solve karke. Upar HARDEST, neeche EASIEST.

Solved Example 1 (HARDEST) - Two dice sum

(Exam favourite)

Do dice ek saath. $P(\text{sum} = 8)$?
Total outcomes = $6 \times 6 = 36$.
Sum 8 wale: $(2,6), (3,5), (4,4), (5,3), (6,2) = 5$.
 $P = 5/36$.

Solved Example 2 - Cards (face card)

(Exam favourite)

52 cards me se 1 nikaali. $P(\text{face card})$?
Face cards = J, Q, K har suit $\rightarrow 3 \times 4 = 12$.
 $P = 12/52 = 3/13$.

Solved Example 3 - Cards (red king)

(Exam favourite)

$P(\text{red king})$? Red kings = 2 (hearts, diamonds).
 $P = 2/52 = 1/26$.

Solved Example 4 - Bag of balls

Bag: 5 red, 3 blue, 2 green (total 10). $P(\text{red})$?
Favourable = 5, total = 10.
 $P = 5/10 = 1/2$.

Solved Example 5 - Complement

$P(E) = 0.3$ hai. $P(\text{not } E)$?
 $P(\text{not } E) = 1 - 0.3 = 0.7$.

Solved Example 6 - Single die

Ek dice. P(even number)?

Even: 2,4,6 -> 3 favourable, total 6.

$$P = 3/6 = 1/2.$$

Solved Example 7 - Coin

Ek coin. P(Head)?

Favourable 1 (H), total 2 (H,T).

$$P = 1/2.$$

Solved Example 8 (EASIEST) - Numbers

1 se 10 tak ek number choose. P(number = 7)?

Favourable 1, total 10.

$$P = 1/10.$$

Q AND A TIME

Q1. CORE: probability ki value kis range mein hoti hai?

Q2. Ek dice. P(number > 4) nikaal.

Q3. 52 cards. P(an ace) nikaal.

Q4. Bag: 4 red, 6 black (total 10). P(black)?

Q5. P(E) = 0.45 hai. P(not E)?

Q6. Do dice. P(sum = 7) nikaal.

Q7. CORE CHECK: ek coin pe Head aane ki probability?

SUMMARY

1. $P(E) = \text{favourable outcomes} / \text{total outcomes}$.

2. $0 \leq P(E) \leq 1$ (hamesha is range mein).

3. $P(\text{not } E) = 1 - P(E)$.

4. Sure event $P=1$, impossible event $P=0$.

5. Total outcomes: coin 2, dice 6, cards 52.

CORE (RED page) revise: experiment/outcome/event,
favourable vs total, fraction, deck of cards, sure/impossible.